

COMPASS_SynopticCB_Collate_DIC_2022_Data

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#Read in and collate all the files in the Processed Data folder

```
files <- list.files(path = "Processed Data", pattern = "\\*.csv$", full.names = TRUE)
```

3. Read and combine all CSVs

```
all_data <- files %>%  
  lapply(read.csv) %>%      # or read.csv if you prefer base R  
  bind_rows()               # combine into one data frame
```

#remove the first unneeded column

```
all_data <- all_data[, -which(names(all_data) == "X")]
```

##Remove sipper samples and clean up that dataframe

##Remove non-Lys Samples

```
all_data_lys <- all_data %>%  
  filter(!str_detect(Sample_ID, "Sip"))
```

#Put sipper samples in their own dataframe

```
all_data_sip <- all_data %>%  
  filter(str_detect(Sample_ID, "Sip"))
```

#clean up the sipper data

```
all_data_sip <- all_data_sip %>%  
  mutate(Year = 2022) %>%  
  mutate(Month = str_extract(Sample_ID, "(?<=2022)\\d{2}")) %>%  
  mutate(  
    Replicate = case_when(  
      str_detect(Sample_ID, "SipA") ~ "A",  
      str_detect(Sample_ID, "SipB") ~ "B",  
      str_detect(Sample_ID, "SipC") ~ "C",  
      TRUE ~ NA_character_ # or "Unknown" if you prefer  
    )  
  ) %>%  
  mutate(  
    Depth_cm = case_when(  
      str_detect(Sample_ID, "10cm") ~ "10",  
      str_detect(Sample_ID, "20cm") ~ "20",  
      str_detect(Sample_ID, "45cm") ~ "45",  
      str_detect(Sample_ID, "SW") ~ "0",  
      TRUE ~ NA_character_ # or "Unknown" if you prefer  
    )  
  )
```

```

    )
  )

#need to pull the dates from the all_data_lys
all_data_lys <- all_data_lys %>%
  mutate(Month = as.numeric(Month))

all_data_sip <- all_data_sip %>%
  mutate(Month = as.numeric(Month))

all_data_sip <- all_data_sip %>%
  mutate(
    Day = all_data_lys$Day[
      match(
        paste(Site, Month),
        paste(all_data_lys$Site, all_data_lys$Month)
      )
    ]
  )
)

```

##Remove and Collate the Sipper Samples

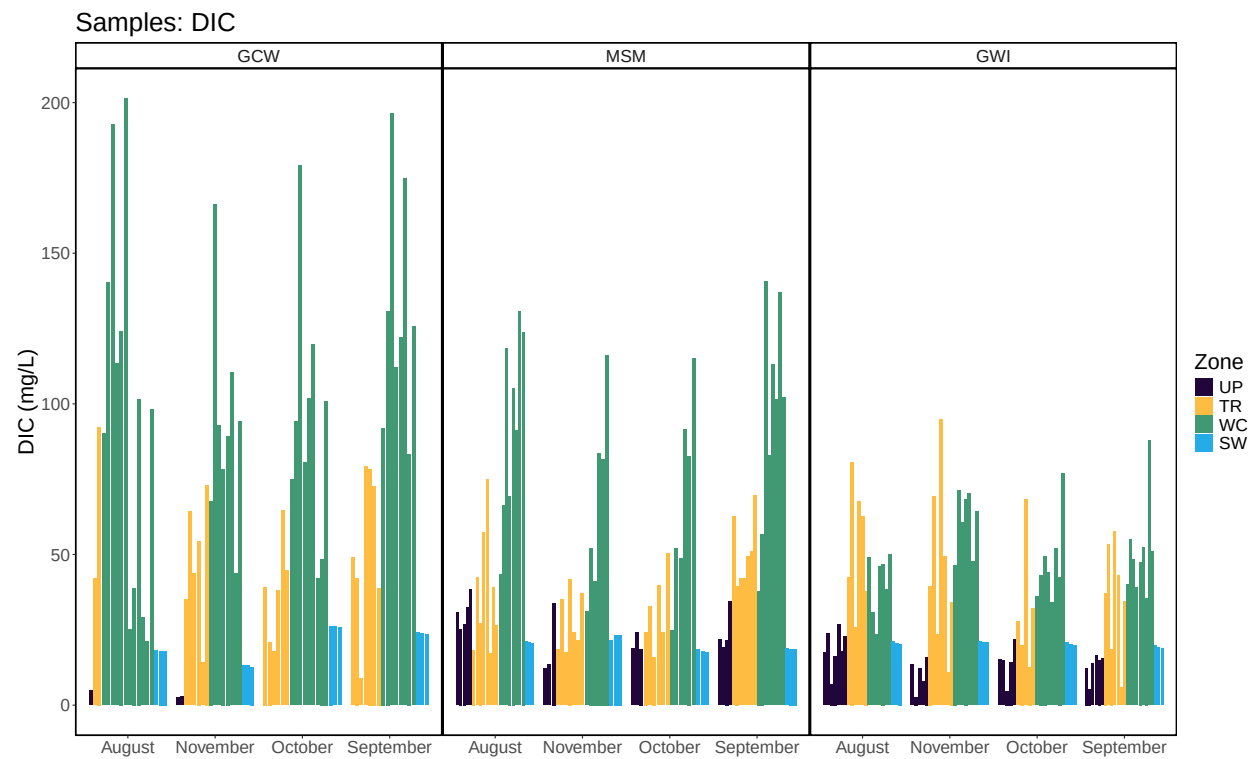
```

#Put together the final lysimeter data
final_data_lys <- all_data_lys %>%
  rename(Analysis_rundate = Analysis_runtime) %>%
  select(
    Project, Region, Site, Zone, Replicate, Depth_cm,
    Sample_ID, Year, Month, Day, Time, Time_Zone,
    ic_mgL, ic_uM, ic_flag,
    Analysis_rundate, Run_notes, Field_notes
    # list columns in the order you want them
  )

#Put together the final sipper data
final_data_sip <- all_data_sip %>%
  rename(Analysis_rundate = Analysis_runtime) %>%
  select(
    Project, Region, Site, Zone, Replicate, Depth_cm,
    Sample_ID, Year, Month, Day, Time, Time_Zone,
    ic_mgL, ic_uM, ic_flag,
    Analysis_rundate, Run_notes, Field_notes
    # list columns in the order you want them
  )
)

```

Visualize Data



Write the 2022 file out to the folder

##END

##Change SWH UPCON to Up and UP to SWAMP

```
all_data_lys <- all_data_lys %>%
  mutate(
    Zone = case_when(
      Site == "SWH" & Zone == "UP" ~ "SWAMP",
      Site == "SWH" & Zone == "UPCON" ~ "UP",
      TRUE ~ Zone # keep everything else unchanged
    )
  )
```